

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 03

DATE : 12-05-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 03

Topics

DATE : 12 - 05 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : Electrostatics [Upto Potential Energy], Friction & Circular Motion

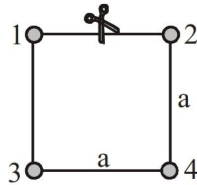
CHEMISTRY : Periodic Table, Thermodynamics, Thermochemistry, Organic 11th,
Reaction MechanismBIOLOGY : Genetics [Till Linkage Map], Plant Diversity (Introduction, Virus, Kingdom
Monera), Human Reproduction [Till Menstrual Cycle], Eye & Ear

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. Four balls, each with mass m , are connected by four nonconducting strings to form a square with side a as shown in figure. The assembly is placed on a horizontal, nonconducting, frictionless surface. Balls 1 and 2 each have charge q , and balls 3 and 4 are uncharged. Find the maximum speed of balls 3 and 4 after the string connecting balls 1 and 2 is cut.



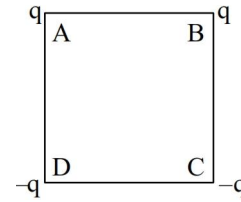
(1) $\sqrt{\frac{q^2}{12\pi\epsilon_0 am}}$

(2) $\sqrt{\frac{q^2}{6\pi\epsilon_0 am}}$

(3) $\sqrt{\frac{q^2}{48\pi\epsilon_0 am}}$

(4) $\sqrt{\frac{q^2}{3\pi\epsilon_0 am}}$

2. Charges are placed on the vertices of a square as $\vec{E}$ be the electric field at the centre. If the charges on A and B are interchanged with those on D and C respectively, then -



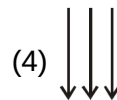
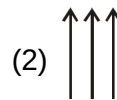
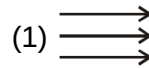
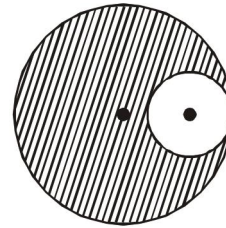
(1) $\vec{E}$ remains unchanged

(2) $\vec{E}$ changes

(3) $\vec{E}$ becomes zero

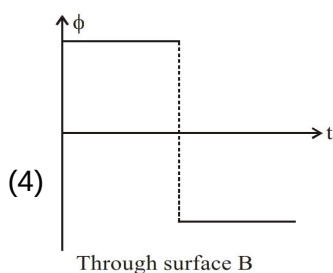
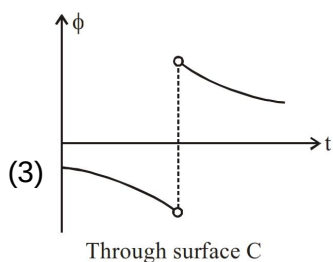
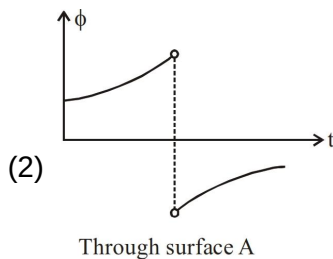
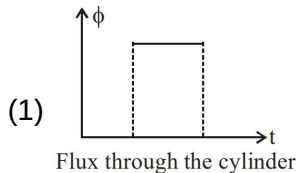
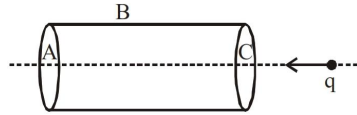
(4) none of these

3. A cavity is made inside a negatively charged dielectric sphere of volume charge density $-\rho$. Which of the following field pattern is correct in the cavity?



SPACE FOR ROUGH WORK

4. Consider a Gaussian surface as shown. A and C are the circular faces while B is the curved surface. A charge q is moved along the axis of cylinder with constant velocity v . Choose the incorrect variation of flux through the surface / cylinder.



5. Gold leaves of an electroscope are negatively charged. When X-rays incident on leaves then leaves :
- (1) Converges
 - (2) Diverges
 - (3) First diverges and then converges
 - (4) Remains at their own position
6. In quark model, a proton and a neutron consist of three quarks each. Two types of quarks, the so called 'up' quark (denoted by u) of charge $+\frac{2}{3}e$ and the 'down' quark (denoted by d) of charge $-\frac{1}{3}e$, together with electrons build up ordinary matter. A possible mark composition of a proton and neutron is :
- (1) udd, udd
 - (2) uud, udd
 - (3) uud, uud
 - (4) ddd, uuu
7. Five balls, numbered 1 to 5 are suspended using separate threads. Pairs (1, 2), (2, 4), (4, 1) show electrostatics attraction, while pairs (2, 3) and (4, 5) show electrostatics repulsion. Therefore, the ball 1 must be
- (1) positively charged.
 - (2) negatively charged.
 - (3) neutral
 - (4) made of metal

SPACE FOR ROUGH WORK

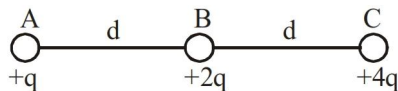
8. Two identical blocks are kept on a frictionless horizontal table and connected by a spring of stiffness 'k' and of natural length ℓ_0 . A total charge Q is distributed on the blocks in a way such that in equilibrium spring elongates by maximum value. If this value is equal to x then value of Q is :

- (1) $2\ell_0\sqrt{4\pi\epsilon_0 k(\ell_0 + x)}$
- (2) $2x\sqrt{4\pi\epsilon_0 k(\ell_0 + x)}$
- (3) $2(\ell_0 + x)\sqrt{4\pi\epsilon_0 kx}$
- (4) $(\ell_0 + x)\sqrt{4\pi\epsilon_0 kx}$

9. The force describing processes occurring in cells of living organisms :-

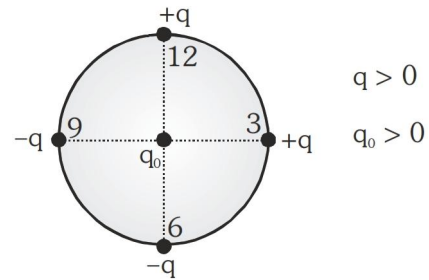
- (1) Weak force
- (2) Strong force
- (3) Nuclear force
- (4) Electro – magnetic force

10. Three charges +q, +2q and +4q are connected by strings as shown in the figure. What is the ratio of tensions in the strings AB and BC.



- (1) 1 : 2
- (2) 1 : 3
- (3) 4 : 1
- (4) 4 : 3

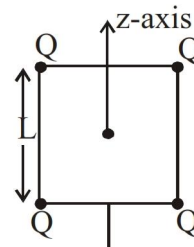
11. Four charges are placed at the circumference of a dial clock as shown in figure.



If the clock has only hour hand then the resultant force on a charge q_0 , placed at the centre points in the direction which shows the time as :

- (1) 1 : 30
- (2) 7 : 30
- (3) 4 : 30
- (4) 10 : 30

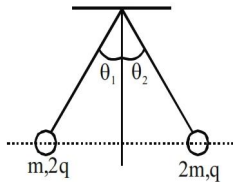
12. Four point +ve charges of same magnitude (Q) are placed at four corners of a rigid square frame in xy plane as shown in figure. The plane of the frame is perpendicular to z-axis. If a -ve point charges is placed at a distance z away from the above frame ($z \ll L$) then:-



- (1) -ve charge oscillates along the z-axis
- (2) it moves away from the frame
- (3) it moves slowly towards the frame and stays in the plane of the frame
- (4) it passes through the frame only once

SPACE FOR ROUGH WORK

13. Two particles which have masses and charges m and $2q$ and $2m$ and q respectively are hung by massless threads from a point as shown in equilibrium particles are on same horizontal line what can be the relation in θ_1 & θ_2 . (θ_1 & θ_2 are not upto scale in figure)



- (1) $\theta_1 > \theta_2$
(2) $\theta_1 < \theta_2$
(3) $\theta_1 = \theta_2$
(4) More information is required.
14. Two identically charged pith balls are suspended from the same point by two massless identical threads density of each ball is ρ . If system is immersed in a medium of density σ , balls remain undeflected, then the dielectric constant of medium is :-

- (1) $\frac{\rho}{\rho - \sigma}$ (2) $\frac{\rho - \sigma}{\rho}$
(3) $\frac{\sigma}{\rho - \sigma}$ (4) $\frac{\rho - \sigma}{\sigma}$

15. A solid sphere of radius R_1 and volume

charge density $\rho = \frac{\rho_0}{r}$ is enclosed by a hollow sphere of radius R_2 with negative surface charge density σ , such that the total charge in the system is zero, ρ_0 is a positive constant and r is the distance from the centre of the sphere. The ratio

$\frac{R_2}{R_1}$ is :-

- (1) $\frac{\sigma}{\rho_0}$ (2) $\sqrt{2\sigma/\rho_0}$
(3) $\sqrt{\rho_0/2\sigma}$ (4) $\frac{\rho_0}{\sigma}$

16. Two point charges Q_1 and Q_2 of equal magnitude are placed at a certain distance from each other. Assuming the field strength to be positive in the positive direction of x-axis

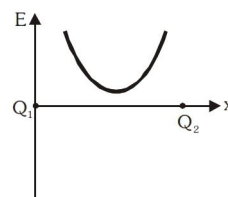


Fig. 1

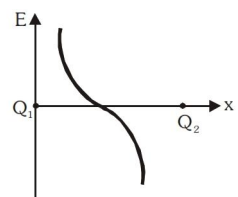


Fig. 2

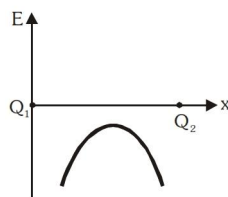


Fig. 3

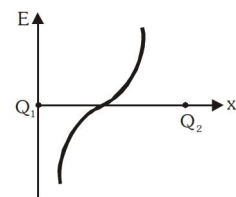


Fig. 4

The signs of the charges Q_1 and Q_2 for given graphs are

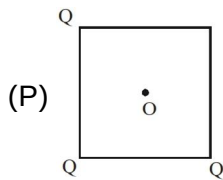
- (1) +, + ; +, - ; -, + ; -, -
(2) +, - ; +, + ; -, + ; -, -
(3) -, + ; +, - ; +, + ; -, -
(4) +, + ; +, - ; -, - ; -, +

SPACE FOR ROUGH WORK

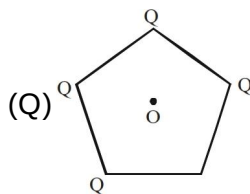
17. In the given diagrams the direction of electric field at point O is given in list-II (O is circumcenter of the given regular polygon). Charge Q is positive. Match the direction of electric field for the given arrangement:

List-I

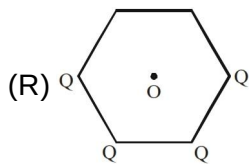
List-II



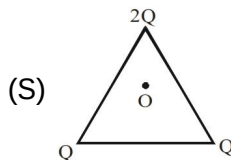
(1)



(2)



(3)

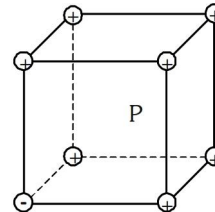


(4)

Code :

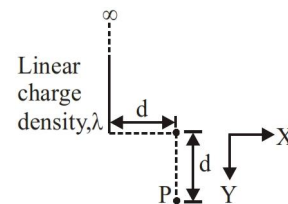
	P	Q	R	S
(1)	2	1	4	3
(2)	3	1	2	4
(3)	3	2	1	4
(4)	1	4	3	2

18. Consider a regular cube with positive point charge +Q in all corners except for one which has a negative point charge -Q. Let the distance from any corner to the center of the cube be r. What is the magnitude of electric field at point P, the center of the cube?



- (1) $E = 0$
 (2) $E = 1 kQ/r^2$
 (3) $E = 2kQ/r^2$
 (4) $E = 6kQ/r^2$

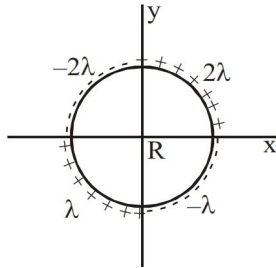
19. Electric field intensity due to a semi-infinite wire at the point P, as shown in the figure given below, is :-



- (1) $\frac{\lambda}{4\pi\epsilon_0 d} \hat{i} + \frac{\lambda}{4\pi\epsilon_0 d} \hat{j}$
 (2) $\frac{\lambda}{4\pi\epsilon_0 d} (1 - \sin \pi/4) \hat{i} + \frac{\lambda}{4\pi\epsilon_0 d} (\cos \pi/4) \hat{j}$
 (3) $\frac{\lambda}{4\pi\epsilon_0 d} (1 - \sin \pi/4) \hat{i} - \frac{\lambda}{4\pi\epsilon_0 d} (\cos \pi/4) \hat{j}$
 (4) $\frac{\lambda}{4\pi\epsilon_0 d} (\sin \pi/4) \hat{i} + \frac{\lambda}{4\pi\epsilon_0 d} (\cos \pi/4) \hat{j}$

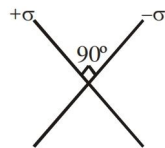
SPACE FOR ROUGH WORK

20. The charge per unit length of the four quadrant of the ring is 2λ , -2λ , λ and $-\lambda$ respectively. The magnitude of electric field at the centre is :



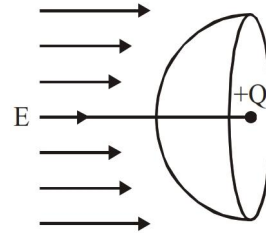
- (1) $\frac{2K\lambda}{r}$ (2) $\frac{K\lambda}{r}$ (3) $\frac{4K\lambda}{r}$ (4) $\frac{3K\lambda}{r}$

21. Two infinite sheets carrying charge density $+\sigma$ and $-\sigma$ arranged perpendicularly as shown in the figure. Select the approximate electric field lines for the charge system near the intersection.



- (1) (2) (3) (4)

22. The figure shows a point charge $+Q$ placed at the center of an imaginary hemispherical surface of radius R . A uniform electric field E is applied in the horizontal direction. If the net flux passing through the hemispherical surface is zero then what is the magnitude of the electric field E .



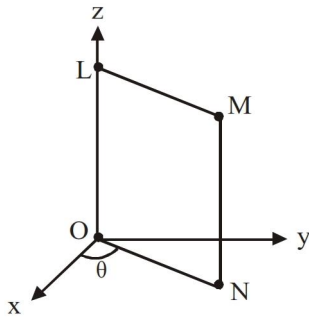
- (1) $\frac{Q}{2\pi\epsilon_0 R^2}$ (2) $\frac{Q}{2\epsilon_0 R^2}$
(3) $\frac{Q}{\pi\epsilon_0 R^2}$ (4) $\frac{Q}{\epsilon_0 R^2}$

23. Consider a spherical symmetric distribution of charge. The centre of symmetry is at O . Electric flux linked with a spherical Gaussian surface of radius ' r ' and centered at O is given by $\phi = \phi_0 r^4$. Volume charge density at a distance ' r ' from O is given by :-

- (1) $\frac{\epsilon_0 \phi_0 r}{\pi}$ (2) $\frac{3\epsilon_0 \phi_0 r}{4}$
(3) $\frac{3\epsilon_0 \phi_0 r}{4\pi}$ (4) $\frac{\epsilon_0 \phi_0 r}{2\pi}$

SPACE FOR ROUGH WORK

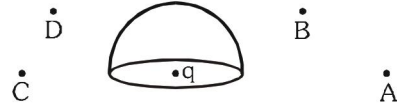
24. The electric field intensity at all points in space is given by $\vec{E} = \sqrt{3}\hat{i} - \hat{j}$ volts/metre. A square frame LMNO of side 1 metre is shown in figure. The point N lies in x-y plane. The initial angle between line ON and x-axis is $\theta = 60^\circ$.



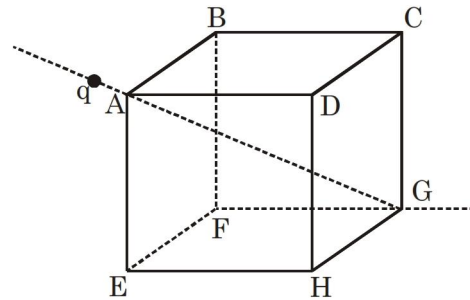
The magnitude of electric flux through area enclosed in square frame LMNO is -

- (1) 0 volt-metre
(2) 1 volt-metre
(3) 2 volt-metre
(4) 4 volt-metre
25. Select incorrect statement :
- (1) Electrostatics lines of forces cannot be closed
(2) Gauss's law is valid for any arbitrary charge distribution
(3) Electrostatic field is a conservative field
(4) Coulomb's law is valid for cylindrical symmetry of charge

26. Figure shows a charge q placed at the centre of a hollow hemisphere. A second charge Q is placed at one of the positions A, B, C and D. In which position (s) of this second charge, the flux of the electric field through the hemisphere remains unchanged ?



- (1) A and C
(2) B and D
(3) A and D
(4) A, B, C and D
27. A charge q is kept just outside a cube on extension of diagonal GA. The magnitude of electric flux of face ABCD is :

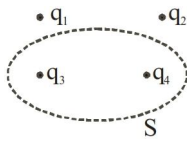


- (1) $\frac{q}{8\epsilon_0}$
(2) $\frac{q}{6\epsilon_0}$
(3) $\frac{q}{24\epsilon_0}$
(4) $\frac{q}{18}\epsilon_0$
28. A point charge $+q$ is kept near infinitely long sheet, the total flux linked with the sheet is :-

- (1) $\frac{q}{\epsilon_0}$
(2) $\frac{q}{\sigma\epsilon_0}$
(3) $\frac{q}{2\epsilon_0}$
(4) Zero

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29. Figure shows four charges q_1, q_2, q_3 and q_4 fixed in space. Then the total flux of electric field through a closed surface S , due to all charges q_1, q_2, q_3 and q_4 is:-

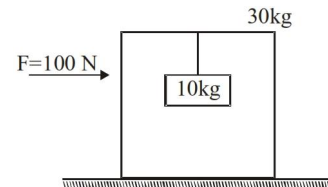


- (1) not equal to the total flux through S due to charges q_3 and q_4
 (2) equal to the total flux through S due to charges q_3 and q_4
 (3) zero if $q_1 + q_2 = q_3 + q_4$
 (4) twice the total flux through S due to charges q_3 and q_4 if $q_1 + q_2 = q_3 + q_4$
30. The person in the drawing is standing on crutches. Assume that the force exerted on each crutch by the ground is directed along the crutch. If the coefficient of static friction between a crutch and the ground is 0.90, determine the largest angle $\theta_{\max}$ that the crutch can have just before it begins to slip on the floor.

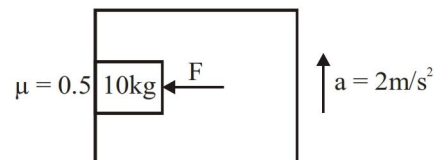


- (1) $\tan^{-1}(0.9)$
 (2) $\cot^{-1}(0.9)$
 (3) $\sin^{-1}(0.9)$
 (4) $\cos^{-1}(0.9)$

31. A force F equals to 100 N is applied on 30 kg box inside which a 10 kg mass is hanging with the help of a massless string as shown in figure. Coefficient of friction between box and surface is 0.1. Initially system is at rest, then at this instant acceleration of box will be



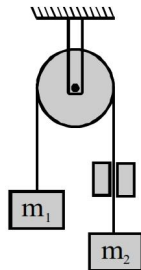
- (1) 4 m/s^2
 (2) 6 m/s^2
 (3) 2 m/s^2
 (4) 1.5 m/s^2
32. A block of mass 10 kg is held on a vertical wall of lift which is going upward with acceleration of 2 m/s^2 . Find minimum value of F so that we can hold the block w.r.t. lift. Friction coefficient between block & wall of lift is 0.5 :-



- (1) 40 N
 (2) 50 N
 (3) 240 N
 (4) 60 N

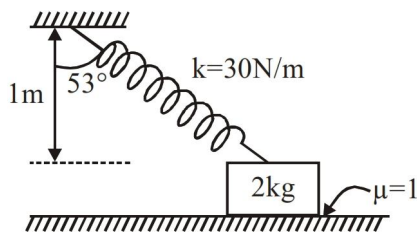
SPACE FOR ROUGH WORK

33. A weightless string passes through a slit over a pulley. The slit offers frictional force f to the string. The string carries two weights having masses m_1 and m_2 where $m_2 > m_1$, then acceleration of the weights is-



- (1) $\frac{(m_2 - m_1)g - f}{m_1 + m_2}$
 (2) $\frac{f - (m_2 - m_1)g}{m_1 + m_2}$
 (3) $\frac{(m_1 + m_2)g - f}{(m_1 - m_2)}$
 (4) $\frac{m_2 g - f}{(m_1 + m_2)}$

34. The spring shown in the figure has a natural length of 1m. What is the initial acceleration of the block when released ?

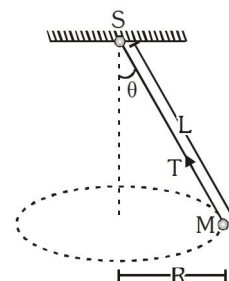


- (1) 4 m/s^2 (2) 6 m/s^2
 (3) 8 m/s^2 (4) 2 m/s^2

35. A circular curve of a highway is designed for traffic moving at 72 km/h. If the radius of the curved path is 100 m, the correct angle of banking of the road should be given by :

- (1) $\tan^{-1} \frac{2}{3}$
 (2) $\tan^{-1} \frac{3}{5}$
 (3) $\tan^{-1} \frac{2}{5}$
 (4) $\tan^{-1} \frac{1}{4}$

36. A string of length L is fixed at one end and carries a mass M at the other end. The string makes $2/\pi$ revolutions per second around the vertical axis through the fixed end as shown in the figure, then tension in the string is



- (1) ML
 (2) $2 ML$
 (3) $4 ML$
 (4) $16 ML$

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37. An aeroplane flying at constant speed 115 m/s towards east, makes a gradual turn following a circular path to fly south. The turn takes 15 seconds to complete. The magnitudes of the centripetal acceleration and average acceleration during the turn, are

(1) $\frac{23\sqrt{2}}{3} \text{ m/s}^2, \frac{23\sqrt{2}}{3} \text{ m/s}^2$

(2) $\frac{46\pi}{3} \text{ m/s}^2, \frac{23\sqrt{2}}{3} \text{ m/s}^2$

(3) $\frac{23\pi}{6} \text{ m/s}^2, \frac{23\sqrt{2}}{3} \text{ m/s}^2$

(4) none of these

38. A car is moving along a circular track with tangential acceleration of magnitude a_0 . It just start to slip at speed v_0 then find radius of circle. (Coefficient of friction is μ) ?

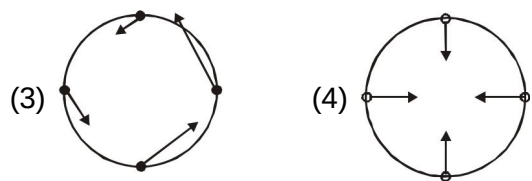
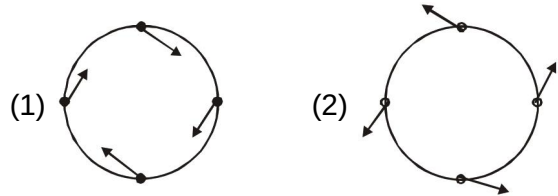
(1) $\frac{v_0^2}{\sqrt{(\mu g)^2 + a_0^2}}$

(2) $\frac{v_0^2}{\sqrt{(\mu g)^2 - a_0^2}}$

(3) $\frac{v_0^2}{\sqrt{(\mu g)^2 - 2a_0^2}}$

(4) $\frac{v_0^2}{\sqrt{(\mu g)^2 + 2a_0^2}}$

39. A car speeds up in a circular path, which of the following figure illustrates the acceleration of the car?



40. A small circular ring has a uniform charge distribution on a far off axial point distance x from the centre of the ring, the electric field is proportional to.

(1) x^{-1}

(2) $x^{-3/2}$

(3) x^{-2}

(4) $x^{5/4}$

41. The electric charge in accelerated motion produces :

(1) an electric field only

(2) a magnetic field only

(3) electromagnetic radiation only

(4) all of the above

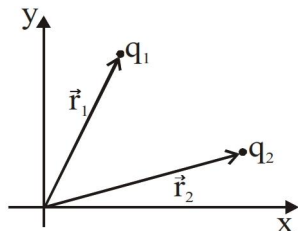
SPACE FOR ROUGH WORK

42. Gauss's law in electrostatics is true, when the charges enclosed in the Gaussian surface are :

- (1) stationary only
- (2) moving only
- (3) moving or stationary
- (4) none of these

43. Two charges q_1 and q_2 are placed in xy plane at positions $\vec{r}_1$ and $\vec{r}_2$ respectively.

Find the $\vec{r}$ where $\vec{E}$ is zero.



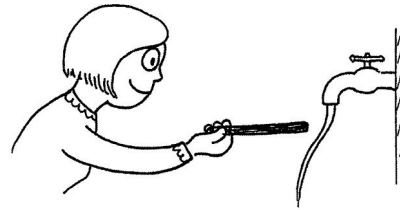
(1) $\frac{\vec{r}_1\sqrt{q_2} + \vec{r}_2\sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}}$

(2) $\frac{\vec{r}_2\sqrt{q_2} + \vec{r}_1\sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}}$

(3) $\frac{\vec{r}_1q_1^2 + \vec{r}_2q_2^2}{q_1^2 + q_2^2}$

(4) $\frac{\vec{r}_2q_1^2 + \vec{r}_1q_2^2}{q_1^2 + q_2^2}$

44. In normal cases thin stream of water bends toward a negatively charged rod. When a positively charged rod is placed near the stream, it will bend in the



- (1) Opposite direction
- (2) Same direction.
- (3) It won't bend at all
- (4) Can't be predicted.

45. A infinitely long non-conducting cylinder of inner radius 5 cm and outer radius 7 cm has uniform volume charge density $\rho = 8.85 \times 10^{-9} \text{ C/m}^2$. The magnitude of electric field at a distance of 10cm from axis of the cylinder is :

- (1) 12 N/C
- (2) 6 N/C
- (3) 18 N/C
- (4) 24 N/C

SPACE FOR ROUGH WORK

46. Which ion has the lowest radius from the following :-

- (1) Na^+ (2) Mg^{2+}
(3) Si^{4+} (4) Al^{3+}

47. If an orbital can accommodate three electrons, then the number of elements that can be placed in the 3rd period of modern periodic table will be :-

- (1) 8 (2) 12
(3) 27 (4) 18

48. The correct order of second ionisation energy is :-

- (1) $\text{Na} > \text{Mg} > \text{Si} > \text{Al}$
(2) $\text{Na} > \text{Mg} > \text{Al} > \text{Si}$
(3) $\text{Mg} < \text{Si} < \text{Al} < \text{Na}$
(4) $\text{Na} < \text{Al} < \text{Mg} < \text{Si}$

49. IE and EA of 'F' are 17.42 and 3.45 eV/atom respectively. EN of F will be :-

- (1) 2.7 (2) 3.7
(3) 4.67 (4) 6.7

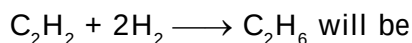
50. Match the column :-

Column-I		Column-II	
(a)	N_2O	(p)	Acidic oxide
(b)	NO_2	(q)	Basic oxide
(c)	Na_2O	(r)	Neutral oxide
(d)	BeO	(s)	Amphoteric oxide

- (1) a – q, b – p, c – s, d – r
(2) a – r, b – p, c – q, d – s
(3) a – r, b – p, c – s, d – q
(4) a – q, b – r, c – s, d – p

51. Given, $\text{BE}_{\text{H-H}} = 420 \text{ kJ mol}^{-1}$,
 $\text{BE}_{\text{C}\equiv\text{C}} = 601 \text{ kJ mol}^{-1}$
 $\text{BE}_{\text{C-C}} = 340 \text{ kJ mol}^{-1}$,
 $\text{BE}_{\text{C-H}} = 425 \text{ kJ mol}^{-1}$

then enthalpy for the reaction,



- (1) -599 kJ mol^{-1}
(2) -580 kJ mol^{-1}
(3) -625 kJ mol^{-1}
(4) -325 kJ mol^{-1}

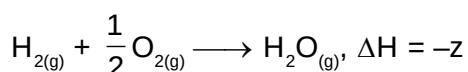
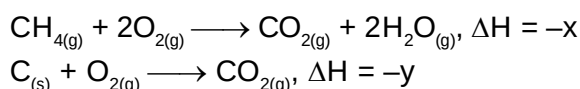
52. The value of ΔH and ΔS for the reaction $\text{C}_{\text{graphite}} + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$ are -100 kJ and -100 J K^{-1} respectively. The reaction will be spontaneous at :-

- (1) 1000 K
(2) 900 K
(3) 1100 K
(4) At any temperature

53. Three moles of an ideal gas expanded into vacuum. Then which is correct?

- (1) $w = 0, \Delta G = 0$
(2) $w = 0, \Delta G < 0$
(3) $w > 0, \Delta G < 0$
(4) $w < 0, \Delta G > 0$

54. The following reactions are given



Then the heat of formation of $\text{CH}_{4(\text{g})}$ will be :-

- (1) $x + y + z$ (2) $y + 2x - z$
(3) $x - y - 2z$ (4) $2z + y - x$

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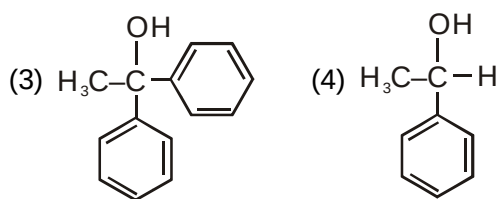
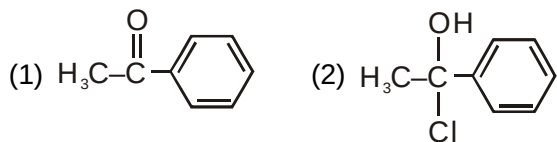
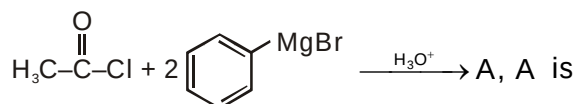
55. The entropy of fusion of water is 5.260 cal/moleK, then enthalpy of fusion of water is :-

- (1) 10.52 kcal/mole
- (2) 0.525 kcal/mole
- (3) 2.225 kcal/mole
- (4) 1.435 kcal/mole

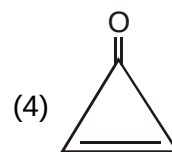
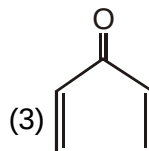
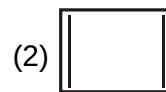
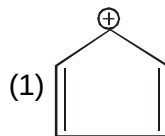
56. Relation between triethyl amine and ethyl methyl propyl amine is :-

- (1) Functional isomers
- (2) Position isomers
- (3) Chain isomers
- (4) Metamers

57. In the reaction



58. The specie which is most stable at room temperature :-

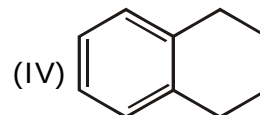
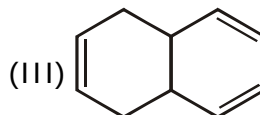
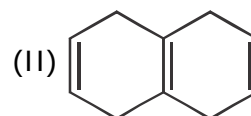
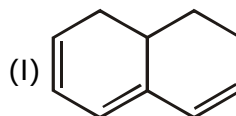


59. How many p-orbitals are participated in the delocalisation of electrons in the given ion?



- (1) 2 (2) 4 (3) 5 (4) 6

60. Give the correct order of magnitudes of heat of hydrogenation of the following :-



- (1) I < IV < III < II (2) IV < III < I < II
(3) IV < I < III < II (4) IV < I < II < III

61. How many structural isomeric alkenes give 2,3-Dimethylpentane on catalytic hydrogenation?

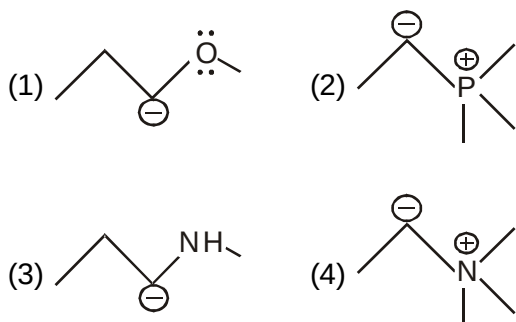
- (1) 2 (2) 3
(3) 4 (4) 5

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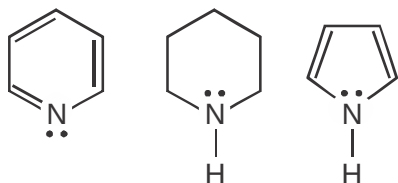
62. The most stable resonating structure is :-

- (1) $\text{H}_2\text{N}-\overset{\oplus}{\text{C}}\text{H}-\text{CH}=\text{CH}-\text{OCH}_3$
- (2) $\text{H}_2\overset{\oplus}{\text{N}}=\text{CH}-\text{CH}=\text{CH}-\text{OCH}_3$
- (3) $\text{H}_2\text{N}-\text{CH}=\text{CH}-\text{CH}=\overset{\oplus}{\text{O}}\text{CH}_3$
- (4) $\text{H}_2\text{N}-\overset{\oplus}{\text{C}}\text{H}-\overset{\ominus}{\text{C}}\text{H}-\text{CH}=\overset{\oplus}{\text{O}}-\text{CH}_3$

63. In which species negative charge is delocalized :-



64. The hybridization states of the nitrogen atoms in pyridine, piperidine and pyrrole are respectively :-

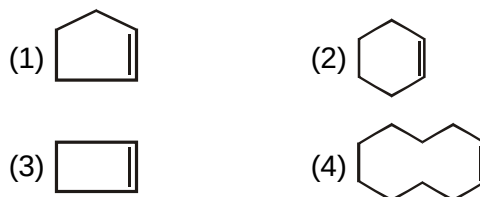


- (1) sp^2 , sp^3 and sp^2
- (2) sp^2 , sp^3 and sp^3
- (3) sp^3 , sp^3 and sp^3
- (4) sp^2 , sp^2 and sp^2

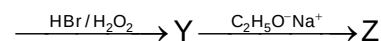
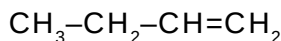
65. IUPAC name of isooctane is :-

- (1) 2,2,3-trimethylpentane
- (2) 2-methylheptane
- (3) 2,2,4-trimethylpentane
- (4) 2,3,4-trimethylpentane

66. Geometrical isomerism is possible in :-



67. Identify Z in the following sequence?

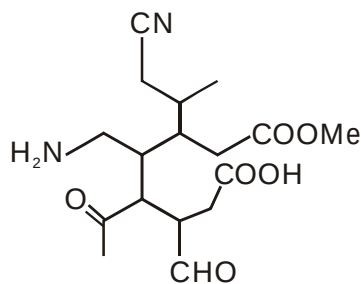


- (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_3$
- (2) $\text{CH}_3-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-\text{O}-\text{CH}_2-\text{CH}_3$
- (3) $\text{CH}_3-(\text{CH}_2)_3-\text{O}-\text{CH}_2-\text{CH}_3$
- (4) $\text{CH}_3-(\text{CH}_2)_4-\text{O}-\text{CH}_3$

68. Which of the following is homologue of $\text{CH}_3\text{NHCH}_2\text{CH}_2\text{CH}_3$

- (1) $\text{CH}_3\text{CH}_2\text{NHCH}_2\text{CH}_3$
- (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
- (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{NH}_2$
- (4) $\text{CH}_3-\text{CH}_2-\text{NH}-\text{CH}_3$

69. Number of functional group and double bond equivalent are respectively.



- (1) 6, 1
- (2) 6, 5
- (3) 6, 6
- (4) 6, 7

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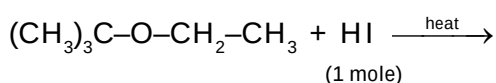
70. Which of the following is correct IUPAC name :-

- (1) 1, 4-dichloro cyclopentane
- (2) 1,4-dichloro-4-ethyl pentane
- (3) 5-bromo-2-chloro hexane
- (4) 5-ethyl-2-methyl hept-1-ene

71. Which of the following behave both as a nucleophile and as an electrophile?

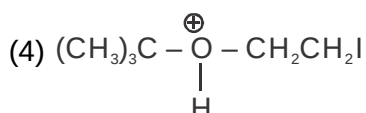
- (1) $\text{CH}_3\text{-C}\equiv\text{N}$
- (2) H_3COH
- (3) $\text{CH}_2=\text{CH-CH}_3$
- (4) $\text{CH}_3\text{-NH}_2$

72. In the reaction,

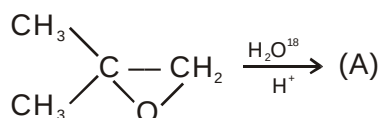


The product(s) formed is(are)

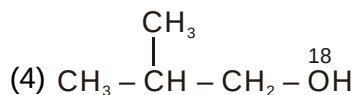
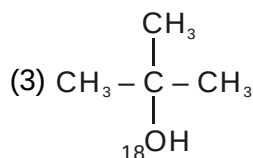
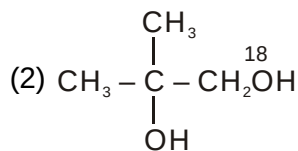
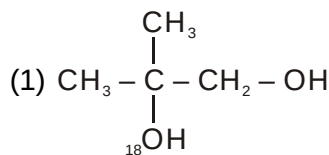
- (1) $(\text{CH}_3)_2\text{C-OH}$ and $\text{CH}_3\text{CH}_2\text{I}$
- (2) $(\text{CH}_3)_3\text{C-I}$ and $\text{CH}_3\text{CH}_2\text{-OH}$
- (3) $(\text{CH}_3)_3\text{C-I}$ and $\text{CH}_3\text{CH}_2\text{I}$



73. In the reaction



The product (A) has the structure :-



74. Vinylic halides are unreactive towards nucleophilic substitution because of the following except :-

- (1) C-halogen bond is strong
- (2) The halogen is bonded to sp^2 carbon.
- (3) A double bond character is developed in carbon - Halogen bond by resonance
- (4) Halide ions are not good leaving group.

75. The correct order of nucleophilicity is :-

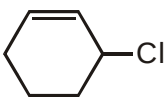
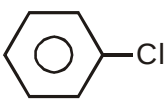
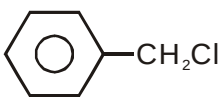
- (1) $\text{CH}_3^- < \text{NH}_2^- < \text{OH}^- < \text{F}^-$
- (2) $\text{F}^- < \text{OH}^- < \text{CH}_3^- < \text{NH}_2^-$
- (3) $\text{OH}^- < \text{NH}_2^- < \text{F}^- < \text{CH}_3^-$
- (4) $\text{F}^- < \text{OH}^- < \text{NH}_2^- < \text{CH}_3^-$

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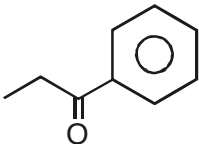
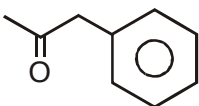
76. For $\text{CH}_3\text{Br} + \text{OH}^- \rightarrow \text{CH}_3\text{OH} + \text{Br}^-$
 The rate of reaction is given by the expression :-

- (1) rate = $k[\text{CH}_3\text{Br}]$
- (2) rate = $k[\text{OH}^-]$
- (3) rate = $k[\text{CH}_3\text{Br}][\text{OH}^-]$
- (4) rate = $k[\text{CH}_3\text{Br}]^\circ[\text{OH}^-]^\circ$

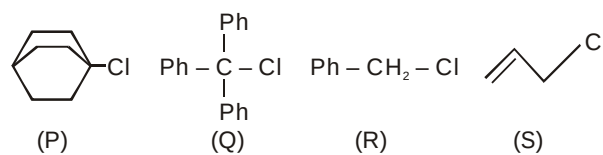
77. Which will give white ppt with AgNO_3 ?

- (1) 
- (2) 
- (3) 
- (4) Both (1) and (3)

78. $\text{C}_6\text{H}_5-\text{C}\equiv\text{C}-\text{CH}_3 \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4, \text{H}_2\text{O}} (\text{A})$ Major Product

- (1) 
- (2) 
- (3) $\text{C}_6\text{H}_5-\text{C}(\text{OH})=\text{CH}-\text{CH}_3$
- (4) $\text{C}_6\text{H}_5-\text{CH}=\text{C}(\text{OH})-\text{CH}_3$

79. Rate of $\text{S}_\text{N}1$ reaction is :-



- (1) $\text{P} > \text{Q} > \text{R} > \text{S}$
- (2) $\text{Q} > \text{R} > \text{S} > \text{P}$
- (3) $\text{P} > \text{R} > \text{Q} > \text{S}$
- (4) $\text{R} > \text{S} > \text{Q} > \text{P}$

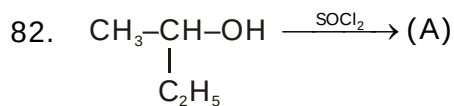
80. Which step in $\text{S}_\text{N}1$ reaction is a rate determining step?

- (1) Attack of nucleophile
- (2) Formation of a racemic mixture
- (3) Formation of a carbocation intermediate
- (4) All of these

81. The reaction of $\text{CH}_3-\text{CH}=\text{CH}-\text{C}_6\text{H}_4-\text{OH}$ with HBr gives

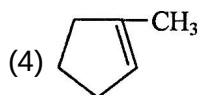
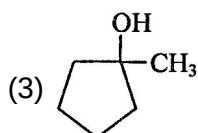
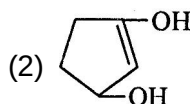
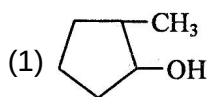
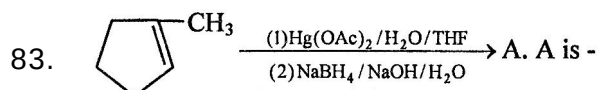
- (1) $\text{CH}_3-\text{CH}=\text{CH}-\text{C}_6\text{H}_4-\text{Br}$
- (2) $\text{CH}_3-\text{CH}_2-\underset{\text{Br}}{\text{CH}}-\text{C}_6\text{H}_4-\text{Br}$
- (3) $\text{CH}_3-\underset{\text{Br}}{\text{CH}}-\text{CH}_2-\text{C}_6\text{H}_4-\text{OH}$
- (4) $\text{CH}_3-\text{CH}_2-\underset{\text{Br}}{\text{CH}}-\text{C}_6\text{H}_4-\text{OH}$

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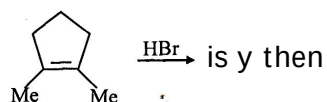
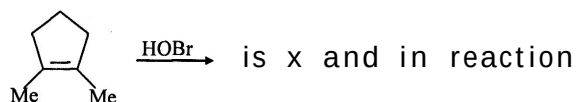


The compound (A) has :-

- (1) Retention of configuration
- (2) inversion of configuration
- (3) racemic mixture
- (4) All of these

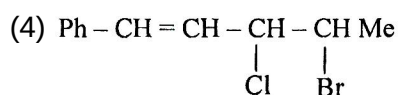
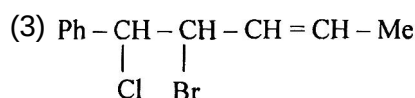
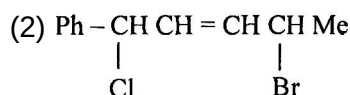
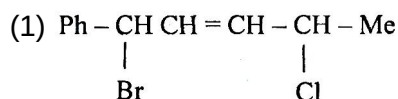
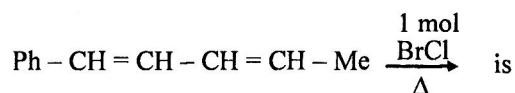


84. If number of products of reaction

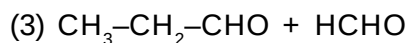
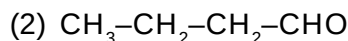
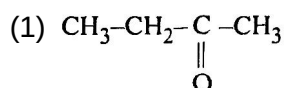


- (1) x = 4, y = 4
- (2) x = 4, y = 2
- (3) x = 2, y = 4
- (4) x = y = 2

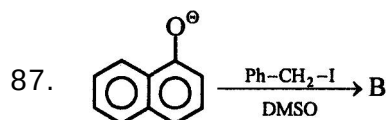
85. The product of reaction



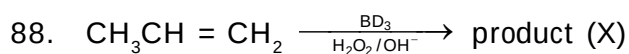
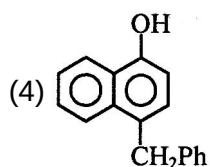
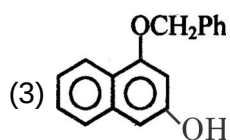
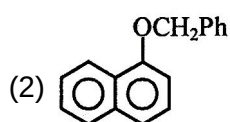
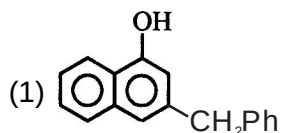
86. The product(s) obtained via oxymercuration ($\text{HgSO}_4 + \text{H}_2\text{SO}_4$) of 1-butyne would be :-



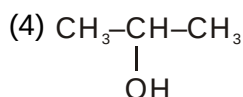
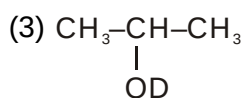
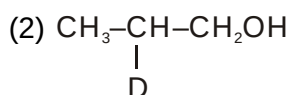
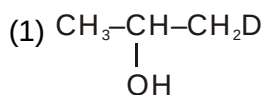
SPACE FOR ROUGH WORK



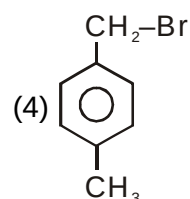
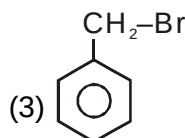
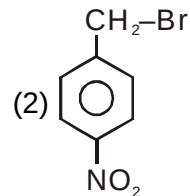
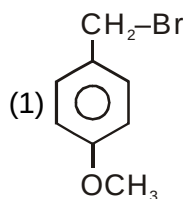
'B' is :-



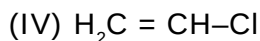
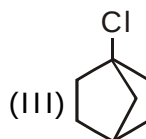
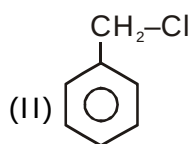
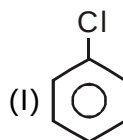
X is :-



89. Which one is more reactive towards $\text{S}_{\text{N}}2$ reaction



90. Which one will be non-reactive towards $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ both :-



(1) I only

(2) I, III, IV only

(3) I, III only

(4) I, II, III, IV all are non-reactive

SPACE FOR ROUGH WORK

91. Gene A and B are linked and are 30 map units apart. A heterozygous individual, whose parents were AAbb and aaBB, would be expected to produce gamete in the following frequencies :-
 (1) 35%-AB, 15%-Ab, 15%-aB, 35%-ab
 (2) 15%-AB, 35%-Ab, 35%-aB, 15%-ab
 (3) 15%-AB, 15%-Ab, 35%-aB, 35%-ab
 (4) 15%-AB, 35%-Ab, 15%-aB, 35%-ab
92. How many types of gametes are produced by a pentahybrid plant (TtAaBbEeGg), in which genes T and A are completely linked?
 (1) 32
 (2) 16
 (3) 8
 (4) can not be determined
93. If 7 alleles are responsible for a character than total possible genotype of this characters will be :-
 (1) 28 (2) 10 (3) 21 (4) 15
94. If blood group of mother is A and father is B than ratio for blood group of offspring A : B : AB : O will be :-
 (1) 1 : 1 : 1 : 1
 (2) 2 : 0 : 2 : 0
 (3) 0 : 0 : 4 : 0
 (4) All of these
95. Which is not related to sickle cell anaemia?
 (1) Codominance
 (2) Lethality
 (3) Pleiotropy
 (4) Multiple allelism
96. Test cross ratio obtained by Bateson & Punnett for two linked genes in trans arrangement was :-
 (1) 1 : 1 : 1 : 1
 (2) 14 (recombinant) : 2 (parental)
 (3) 7 : 1 : 1 : 7
 (4) 1 : 7 : 7 : 1
97. A Snapdragon plant with golden leaf colour and pink flower is self pollinated then what would be phenotypic ratio?
 (1) 3 : 6 : 3 : 1 : 2 : 1
 (2) 9 : 3 : 3 : 1
 (3) 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1
 (4) 2 : 4 : 2 : 1 : 2 : 1
98. In a plant two independently assorting genes C and P are necessary for the production of coloured flowers. In the absence of any one of the gene's dominant allele flower colour becomes white. A plant with genotype ccPP is selfed, what phenotypic ratio will be observed?
 (1) 9 : 7 (2) 1 : 1 : 1 : 1
 (3) 9 : 3 : 3 : 1 (4) All white
99. Sickle-Cell Anaemia is an autosomal recessive disorder. The effect is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of β globin chain of haemoglobin molecule. This substitution results due to single base substitution of DNA from
 (1) - GAG- to- GUG-
 (2) -CTC-to-CAC-
 (3) - CAC - to -CTC-
 (4) -GUG - to - GAG
100. Identify the correct match from the columns I, II and III :
- | Column-I | | Column-II | | Column-III | |
|----------|----------------------|-----------|-------------|------------|-------|
| (1) | Complete dominance | (a) | Non allelic | (i) | 1:2:1 |
| (2) | Incomplete dominance | (b) | Allelic | (ii) | 3:1 |
| (3) | Complementary gene | (c) | Non Allelic | (iii) | 9:3:4 |
| (4) | Supplementary gene | (d) | Allelic | (iv) | 9:7 |
- (1) 1-b-i, 2-d-ii, 3-a-iii, 4-c-iv
 (2) 1-b-ii, 2-d-i, 3-a-iv, 4-c-iii
 (3) 1-c-ii, 2-a-i, 3-b-iv, 4-c-iii
 (4) 1-c-i, 2-a-ii, 3-b-iii, 4-d-iv

101. Garden pea has 7 pair of chromosome. How many linkage groups will be present ?
(1) 14 (2) 1 (3) 7 (4) 28
102. Coupling and repulsion hypothesis of linkage was given by
(1) Bateson and Punnet
(2) Morgan
(3) Sturtevant
(4) Correns
103. The recombination frequency between gene A and B is 15%. What is the distance between gene A and B ?
(1) 5 cM
(2) 15 cM
(3) 10 cM
(4) Cannot determined
104. Number of linkage groups present in human male and female are respectively :
(1) 23, 24 (2) 24, 23
(3) 23, 23 (4) 24, 22
105. Distance between the genes a, b, c and d in map units is $a-d = 3.5$, $b-c = 1$, $a-b = 6$, $c-d = 1.5$ and $a-c = 5$. Find out the sequence of the genes.
(1) a d c b (2) a c d b
(3) a b c d (4) a c b d
106. Which of the following is not an example of mitochondrial inheritance?
(1) Poky form in Neurospora
(2) Petite form in yeast
(3) Ioap inheritance in maize
(4) Male sterility in maize
107. Which of the following statement is true about test cross :-
(a) Crossing the F_1 hybrid with a double recessive genotype
(b) To determine whether F_1 plant is homozygous or heterozygous
(c) Cross between a homozygous dominant and heterozygous dominant plant
(d) Same phenotypic and genotypic ratio
(1) a and c only (2) b and c only
(3) a, b only (4) All except c
108. The genes of male sterility & albinism in plants, are found respectively in :-
(1) Mitochondria & Plastid
(2) Plastid & Mitochondria
(3) Plastid & Plastid
(4) None of the above
109. What will be the genotypic ratio of offsprings of F_2 generation obtained by selfing of F_1 plant having AABbCC genotype
(1) 1 : 2 : 1
(2) 1 : 1
(3) 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1
(4) 27 : 9 : 9 : 9 : 3 : 3 : 3 : 1
110. Read the following statements about hybridization technique of mendel and select the correct statement:-
(A) Emasculation of both male and female parents is required.
(B) Emasculation of only male parent is required.
(C) Emasculation of female parent is required.
(D) Emasculation is not necessary.
(1) A
(2) B
(3) C
(4) All correct
111. A cross is made between $AaBB \times AABb$ genotype individuals, then find out the correct option of probabilities of following offsprings

	A AABB	B AaBB	C AaBb	D aabb
(1)	1/4	1/4	1/4	0
(2)	1/8	1/8	4/16	0/16
(3)	1/16	2/16	2/8	1/16
(4)	2/8	2/8	4/8	0/16

112. In pea plant, genotype (Bb) shows -
 (1) Round seed with large sized starch grain
 (2) Round seed with intermediate sized starch grain
 (3) Round seed with small sized starch grain
 (4) Wrinkled seed with small sized starch
113. A non-inheritable environmentally induced modification resembling the phenotype of a known gene, refers to :-
 (1) Phenocopy (2) Genocopy
 (3) Genotype (4) Phenotype
114. "When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters". This explains :-
 (1) Law of dominance
 (2) Law of segregation
 (3) Law of independent assortment
 (4) Postulate of paired factors
115. In a diploid organism how many maximum different alleles of a gene are possible :-
 (1) One
 (2) Two
 (3) Many
 (4) Variable in different chromosome
116. How many kinds of gametes can be produced by an individual of genotype MmNneeAaBB.
 (1) 32 (2) 16 (3) 4 (4) 8
117. Which of the following type of gamete will not be produced normally by genotype AaBbCc ?
 (1) AbC (2) ABc
 (3) aBC (4) aBb
118. What will be probability of non parental offsprings in the F_2 generation of a quantitative character regulated by two genes :-
 (1) $\frac{4}{16}$ (2) $\frac{1}{8}$ (3) $\frac{6}{16}$ (4) $\frac{14}{16}$

119. Which one of the following characters chosen by Mendel can express in heterozygous condition ?
 (1) Green seed colour
 (2) Constricted pod shape
 (3) Terminal flower position
 (4) Green pod colour
120. The Punnett square shown below represents the pattern of inheritance in dihybrid cross, when round (R) is dominant over wrinkled (r) seeds and yellow (Y) is dominant over green (y) seeds.

	RY	Ry	rY	ry
RY	F	J	N	R
Ry	G	K	O	S
rY	H	L	P	T
ry	I	M	Q	U

Find out the odd one –

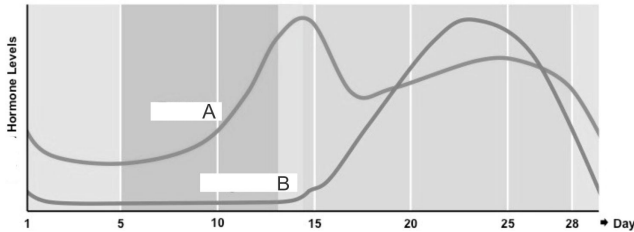
- (1) G (2) K (3) H (4) L
121. Which statements is not correct ?
 (1) Gametes are always hemizygous.
 (2) Detectable variant of character is called as traits.
 (3) Unit of Heredity is gene.
 (4) Chemical unit of Heredity is DNA.
122. Complementary and Epistasis gene interaction ratio is respectively -
 (1) 9 : 3 : 4, 9 : 7
 (2) 9 : 7, 12 : 3 : 1
 (3) 9 : 3 : 3 : 1, 9 : 7
 (4) 12 : 3 : 1, 9 : 7
123. Lathyrus odoratus is an example of which of the following genes :-
 (1) Supplementary genes
 (2) Complementary genes
 (3) Lethal genes
 (4) Codominant genes

124. A dihybrid cross produces 6 phenotypes in F_2 generation then which statement is correct :-
- Both gene show dominance
 - Both gene show incomplete dominance
 - One is dominance and another is incomplete dominance
 - Both gene show codominance
125. Plant with genotype AaBbCc crossed with AABbCC then find out probability of plant with genotype AABbCc
- $\frac{2}{64}$
 - $\frac{4}{64}$
 - $\frac{1}{8}$
 - $\frac{3}{8}$
126. Which of the following have bacteriorhodopsin?
- Chlamydia
 - Rickettsia
 - Mycoplasma
 - Halophiles
127. Select the mismatch :
- Clostridium* - Mycorrhiza
 - Anabaena* - Azolla
 - Aulosira* - Paddy field
 - Frankia* - *Alnus*
128. Nucleoid of Bacteria possess
- SS DNA + SS RNA + Acidic proteins
 - DS DNA + SS RNA + Histone proteins
 - DS DNA + SS RNA + Polyamines
 - DS DNA + SS RNA + Acidic proteins
129. *E. Coli* and *Anabaena* resemble in the presence of :-
- Heterocyst
 - Oxygenic photosynthesis
 - Peptidoglycan cell wall
 - Chromatophore
130. Read the following statements w.r.t. heterocyst and select the correct option :-
- Thickened cell wall
 - They lack PS-II
 - Nitrogen fixing structure
 - Site for oxygen evolution
 - Help in gene recombination
- Only (C) is correct
 - Only (E) is incorrect
 - (D) and (E) are incorrect
 - All are correct
131. Which of the following organism satisfy the criteria of simple in structure and complex in behaviour?
- Euglena*, *Chlamydomonas*, *Chlorella*
 - Triticum*, *Mangifera*, *Oryza*
 - Bacillus*, *Salmonella*, *Xanthomonas*
 - Volvox*, *Fucus*, *Chara*
132. Which of the following statement is correct for Eubacteria?
- Mesosomes helps in septa formation during binary fission
 - Rod shaped bacteria are called *Bacillus*
 - Teichoic acid present in the cell wall of gram positive bacteria
 - All of these
133. Which of the following is a retro virus?
- HIV
 - Herpes virus
 - Pox virus
 - Variola virus
134. In which of the following group of prokaryotes photolysis of water occur during photosynthesis?
- Rhodospseudomonas*, *Rhodospirillum*
 - Nitromonas*, *Nitrococcus*
 - Nostoc*, *Anabaena*
 - Rhizobium*, *Frankia*
135. Mycoplasma is not affected by penicillin due to :-
- Pleomorphic nature
 - Presence of linear DNA
 - Lack of cell wall
 - Presence of peptidoglycan cell wall

136. Scrotum is connected with the abdomen by

- (1) Detrusor muscles
- (2) Dartos muscles
- (3) Gubernaculum
- (4) Inguinal canal

137. Identify the hormones labelled as A and B in the following graph.

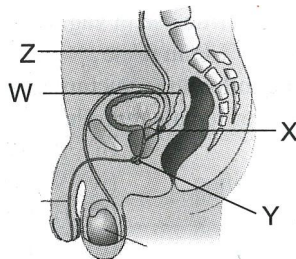


- (1) A - LH, B - Progesterone
- (2) A - Oestrogen, B - FSH
- (3) A - Oestrogen, B - Progesterone
- (4) A - Progesterone, B - Oestrogen

138. The muscles present in the wall of scrotum is

- (1) Detrusor
- (2) Cremaster
- (3) Dartos
- (4) Intercostal

139. Which among the following releases secretions that helps in lubrication of the penis?



- (1) Z
- (2) W
- (3) X
- (4) Y

140. The number of mammary lobes present in each breast of human female is

- (1) 15 – 20
- (2) 5 – 10
- (3) 30 – 40
- (4) 50 – 70

141. In female external genitalia, a pair of large skin folds homologous with scrotum of male is -

- (1) Labia minora
- (2) Labia majora
- (3) Clitoris
- (4) Fourchette

142. In mammary glands milk is sucked out through

- (1) Ampulla
- (2) Alveoli
- (3) Lactiferous duct
- (4) Mammary ductules

143. Which of the following structure is/are present in vestibule part of female reproductive system?

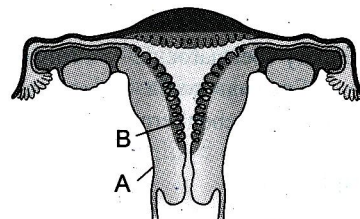
- (i) Clitoris
- (ii) Perineum
- (iii) Urethral orifice
- (iv) Vaginal orifice

- (1) (i) and (ii)
- (2) (iii) and (iv)
- (3) (ii) and (iii)
- (4) (i) and (iv)

144. The cervical canal opens into vagina by an opening called

- (1) Internal Os
- (2) External Os
- (3) Ampulla
- (4) Fundus

145. Select the option which gives correct identification of labelled parts A and B along with functional characteristics in the given sectional view of female reproductive system.



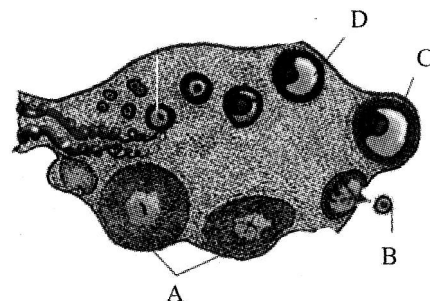
- (1) A - perimetrium, composed of ciliated cuboidal epithelium
- (2) B - Myometrium, which undergoes strong uterine contraction at the time of child birth
- (3) B - Endometrium is the inner glandular lining formed of columnar epithelium
- (4) A - Myometrium, made of smooth muscles

146. Which of the following hormone controls the process of spermiogenesis?

- (1) LH
- (2) ICSH
- (3) FSH
- (3) GnRH

147. How many polar bodies are formed during gametogenesis in males?
 (1) 2 (2) 1 (3) 3 (4) none
148. For normal fertility in male, at least
 (1) 60% sperms must have normal shape and size while rest 40% sperms have high motility
 (2) 40% sperms should have normal shape, size and motility
 (3) 40% sperms have normal shape and size while rest 60% have high motility
 (4) 60% sperms must have normal shape and size while 40% of them should show vigorous motility
149. How many spermatozoa will be produced by 200 primary spermatocytes and how many ova are produced by 100 primary oocytes?
 (1) 400 sperms + 200 ova
 (2) 200 sperms + 100 ova
 (3) 200 sperms + 200 ova
 (4) 800 sperms + 100 ova
150. Which of the following is the correct sequence in the process of spermatogenesis?
 (1) Spermatogonia → Spermatids → Primary spermatocyte → Secondary spermatocyte → Spermatozoa
 (2) Spermatogonia → Primary spermatocyte → Spermatid → Secondary spermatocyte
 (3) Spermatogonia → Primary spermatocyte → Secondary spermatocyte → spermatids → Spermatozoa
 (4) Secondary spermatocyte → Primary spermatocyte → Spermatogonia → Spermatid
151. In human sperm, Nebenkern refers to
 (1) Acrosome
 (2) Axial filament
 (3) Proximal centriole
 (4) Spiral Mitochondria
152. Difference between oogenesis and spermatogenesis is:
 (1) Spermatogenesis is not completed until after fertilization occurs, but oogenesis is completed at the time of birth
 (2) Oogenesis begins at puberty and spermatogenesis begins during embryonic life
 (3) Primary oocyte forms four haploid ova while primary spermatocyte forms only one haploid sperm
 (4) Cytoplasmic division is unequal in oogenesis and equal in spermatogenesis
153. Spermateliosis is another name of
 (1) Spermatogenesis
 (2) Spermiation
 (3) Spermiogenesis
 (4) Degeneration of sperms
154. If in a sperm a distal centriole is removed, which of the following shall not occur?
 (1) Acrosomal reaction
 (2) Movement of sperm
 (3) First cleavage of zygote
 (4) Amphimixis
155. Find the odd statement.
 (1) Human male ejaculates about 200 - 300 million sperms during a coitus.
 (2) Type A spermatogonia serves as stem cell for spermatogenic lineage.
 (3) Spermiogenesis refers to maturation of sperms from their precursor spermatids.
 (4) Spermatids have 44 + XY chromosomes only.
156. Menarche is the stage when
 (1) First ejaculation occurs
 (2) First menstrual cycle occurs
 (3) Hymen breaks down
 (4) Parturition takes place

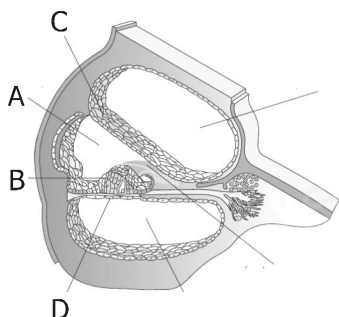
157. Stroma of the mammalian ovary consists of
- Connective tissue
 - Nerves
 - Vascular tissue
 - All of these
158. Corpus luteum is developed from
- Primary follicle
 - Ovarian stroma
 - Left over follicle after release of ovum
 - Ovum
159. Zona pellucida, a glycoprotein covering in mature follicle, is an example of
- Primary egg membrane
 - Secondary egg membrane
 - Tertiary egg membrane
 - None of these
160. Which of the following statement is incorrect about menstruation?
- Around 40 – 80 ml blood loss takes places.
 - Menses take place for 3 – 5 days.
 - Beginning of menstrual cycle at puberty is called menarche.
 - Menstrual fluid can clot easily.
161. The levels of which of the following hormone rises in urine, in females after menopause?
- Oestrogen
 - Progesterone
 - FSH
 - Androgens
162. Estrous cycle is characteristic of
- Human females
 - All vertebrates
 - All mammals
 - Non-primate mammalian females
163. Which of the following is also called 'yellow body'?
- Corpus albicans
 - Corpus luteum
 - Graafian follicle
 - Cumulus oophoricus
164. Lack of menstruation may be indicative of
- Stress
 - Poor health
 - Pregnancy
 - All of these
165. If the menstrual cycle of a woman is of 25 days, then ovulation occurs on
- 11th day
 - 14th day
 - 20th day
 - 7th day
166. Mark the stage formed just after release of secondary oocyte (ovulation) due to rupturing of Graafian follicle.
- Corpus luteum
 - Corpus albicans
 - Corpus epididymis
 - Corpus haemorrhagicum
167. During ovulation in human female
- Peak of oestrogen is higher than FSH
 - Peak of FSH is higher than LH
 - Peak of LH is higher than FSH
 - Tallest peak of progesterone is observed
168. Antrum, a fluid-filled cavity is typically seen in which follicle?
- Primordial follicle
 - Primary follicle
 - Secondary follicle
 - Graafian follicle
169. The diagram of the sectional view of the ovary is given below. The labelled structure that secretes large amount of progesterone is



- A
- B
- C
- D

170. Second polar body is formed inside
- Ovary
 - Fallopian Tube
 - Vagina
 - Uterus

171. Identify the parts labelled A, B, C and D in the figure and match the correct names from the list (i–viii) given



- (i) Reissner's membrane
- (ii) Basilar membrane
- (iii) Tectorial membrane
- (iv) Organ of corti
- (v) Hair cells
- (vi) Otolith organ
- (vii) Scala media
- (viii) Scala vestibuli

A	B	C	D
(1) (ii)	(v)	(iii)	(i)
(2) (vii)	(iv)	(ii)	(iii)
(3) (vii)	(iv)	(i)	(ii)
(4) (viii)	(vi)	(i)	(iii)

172. The part of ear where sound is transduced in impulse form is :

- (1) Tympanic membrane
- (2) Ear ossicles
- (3) Semicircular canals
- (4) Cochlea

173. A small passage that permits continuity between scala vestibuli and scala tympani :

- (1) Helicotrema
- (2) Eustachian tube
- (3) Cochlea
- (4) Vestibule

174. The gelatinous membrane covering the sensory hair cells of the ear is known as :

- (1) Reissner's membrane
- (2) Tectorial membrane
- (3) Basilar membrane
- (4) Neuro-Sensory membrane

175. Which of following is a correct match of ear part and its function :

- (1) Organ of corti—increases the efficiency of sound waves
- (2) Eustachian tube – Maintain body balance and posture
- (3) Tectorial membrane – Determines patterns of vibration of sound waves
- (4) Semicircular canal – Equalises the pressure on either sides of the ear drum

176. What is correct about blind spot of eye.

- (1) Optical nerve leave from it
- (2) Blood vessels enter to it
- (3) Photoreceptors cells are not present in this region
- (4) All are correct

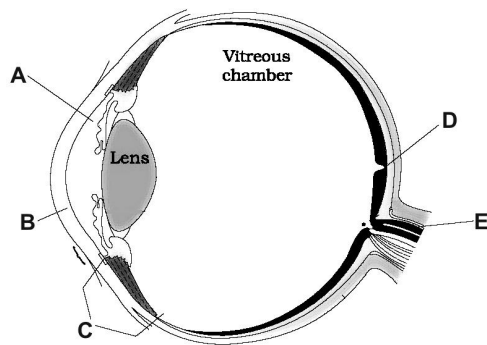
177. Select wrong statement :

- (1) At the posterior pole of the eye lateral to blind spot, a yellowish pigmented spot is present called macula lutea and its central pitted part is called fovea.
- (2) Fovea is out thinned portion of retina
- (3) Fovea contain only cones, no rods are present and resolution is the greatest
- (4) Aqueous chamber is space between lens and cornea and vitreous chamber is space between iris & pupil.

178. Inner layer eye retina contains three layers of cells from outside to inside are :

- (1) Ganglion cells → Bipolar → Photo-receptor cells
- (2) Photo-receptor cells → Bipolar → Ganglion cells
- (3) Bipolar → Ganglion cells → Photo-receptor cells
- (4) Photo-receptor cells → Ganglion cells → Bipolar

179. Identify A, B, C, D in given eye diagram :



- (1) A = Cornea, B = Aqueous chamber,
C = Fovea, D = Ciliary body,
E = Optic nerve
- (2) A = Optic nerve, B = Cornea,
C = Ciliary body, D = Fovea,
E = Aqueous chamber
- (3) A = Fovea, B = Aqueous chamber,
C = Cornea, D = Ciliary body,
E = Optic nerve
- (4) A = Aqueous chamber, B = Cornea,
C = Ciliary body, D = Fovea,
E = Optic nerve

180. Match the column :

- | | |
|---------------|------------------------------|
| (a) Day light | (i) An aldehyde of vitamin-A |
| (b) Twilight | (ii) Photopic |
| (c) Rhodopsin | (iii) Scotopic |
| (d) Retinal | (iv) Visual purple |

(1) a = ii, b = iii, c = iv, d = i

(2) a = iii, b = i, c = iv, d = ii

(3) a = iv, b = ii, c = i, d = iii

(4) a = i, b = ii, c = iii, d = iv

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